Math 126 Assignment 5

1. Solve the initial value problem

$$9\partial_t^2 u - \partial_x^2 u = 0, \quad u(0,x) = x^2, \quad \partial_t u(0,x) = e^x - x.$$

2. Find conditions on h for which the problem

$$(\partial_t^2 - \partial_x^2)u(t,x) = 0, \quad u(0,x) = 0, \; \partial_t u(0,x) = h(x), \quad 0 < x < L,$$
$$\partial_x u(t,0) = \partial_x u(t,L) = 0, \quad t \geq 0,$$

has a solution $u \in C^2$. Write a formula for that solution.

3. Suppose that

$$\partial_t^2 u - \partial_x^2 u = 0$$

and let (t_j, x_j), $j = 1,2,3,4$ be the vertices of a parallelogram whose sides are given by characteristics $x \pm t = \text{const}$. Show that $u(t_j, x_j)$ is determined by $\{u(t_\ell, x_\ell)\}_{\ell \neq j}$.